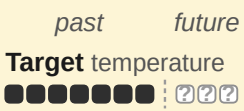
How Kairos makes a forecast

ShanghaiTech University and Ant Group · 50M parameters · encoder-decoder with adaptive patch sizes · zero-shot

1. One series at a time

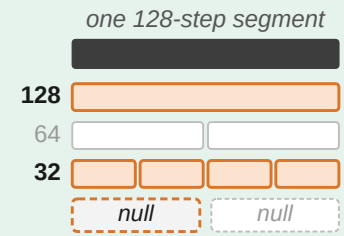
Kairos reads only the target's own past, up to 2,048 steps. A ? is a value not known yet.



No covariates; each metal on its own.

2. Patch size per segment

The scaled context is cut into 128-step segments. For each one a router keeps 3 of 5 options: patches of 128, 64 or 32 steps, or a null that adds nothing.

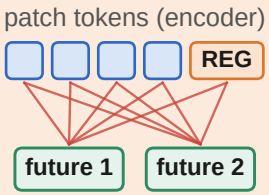


router keeps 3 of 5 per segment

Busy stretches can get fine patches and smooth ones coarse patches; the kept sizes are blended into tokens at the finest kept size.

3. Encoder, then a decoder

Six encoder blocks read the patch tokens. Six decoder blocks start from two learned forecast tokens and read the encoder through cross-attention.



two learned forecast tokens

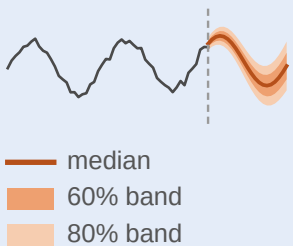
Positions (DRoPE): rotary, with frequencies tuned to each series' spectrum

Cross-attention: forecast tokens read the encoder

Feed forward: 512 → 2,048 → 512, ReLU

4. 128 steps in one pass

A residual MLP turns each forecast token into **64 steps × 9 quantiles**, so one pass covers 128 steps.



Our runs also average with the forecast of −x and clip at zero a series that was never negative. The median is the point forecast.

Kairos lets each series choose how finely it is cut into tokens, segment by segment.

6 + 6 blocks · width 512 · 8 heads · patch sizes 128, 64, 32 and 2 nulls, top 3 · 2 forecast tokens of 64 steps

In the notebooks: last 2,048 steps · each metal on its own · no covariates · every horizon in one pass